

Started on	Tuesday, 19 January 2021, 4:00 PM
State	Finished
Completed on	Tuesday, 19 January 2021, 5:59 PM
Time taken	1 hour 59 mins
Grade	Not yet graded

Question **1**

Correct

Mark 1.00 out of 1.00

Valuator type interaction technique could be:

Select one or more:

- ☒ a. Slider ✓
- ☐ b. Check boxes
- ☐ c. Menu
- ☒ d. Scroll bar ✓

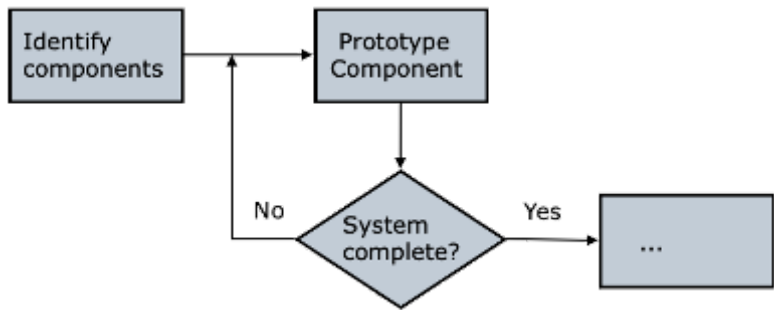
Your answer is correct.

Question **2**

Correct

Mark 1.00 out of 1.00

The figure describes the prototyping approach by:



Select one:

- ☐ a. Revolutionary prototyping
- ☐ b. Vertical prototyping
- ☒ c. Incremental prototyping ✓
- ☐ d. Evolutionary prototyping

Your answer is correct.

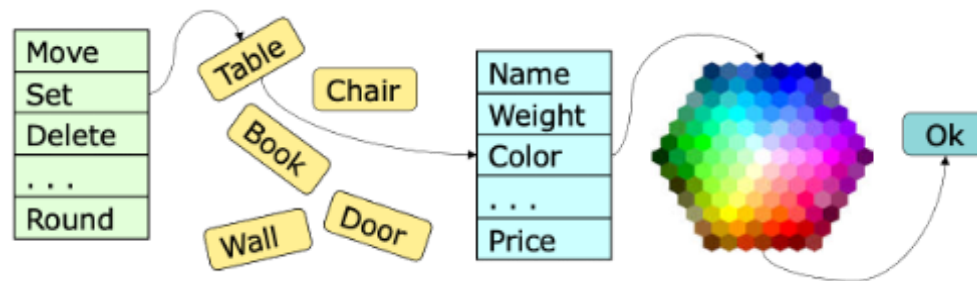
Question **3**

Complete

Marked out of 5.00

Exemplify and explain the using of the concept of Dialogue Cell in the graphical interface in the figure, to execute the following command: Set Table Color to RED <ok>

Highlight and detail the issues related with the elements of the dialogue cell.



Dialogue independence can change the interface design without affecting the code from the application. This develops alternative user interfaces for the same application (semantics). It also facilitates the iterative refinement of the interface using prototyping and testing.

A dialogue cell describes a transaction between user and system. The dialogue cell system allows a programmer to specify complex graphical user interfaces in a rather easy way using a special language, a compiler and a run time system.

Using input gathering(=information that says how to group the actions) we choose the task "set". By doing this, we implicit set the start.

set --> table --> color -(to)-> red --> ok

The 'OK' button calls the start of the app. It invokes the method 'SET' and starts the execution.

The syntactic form: *Set Table Color to Red^

The user can build the syntactic by:

- GUI (Graphical user interface) -> it selects directly the objects
- Voice interface
- Linguistic interface

The exemplification is shown in the picture.

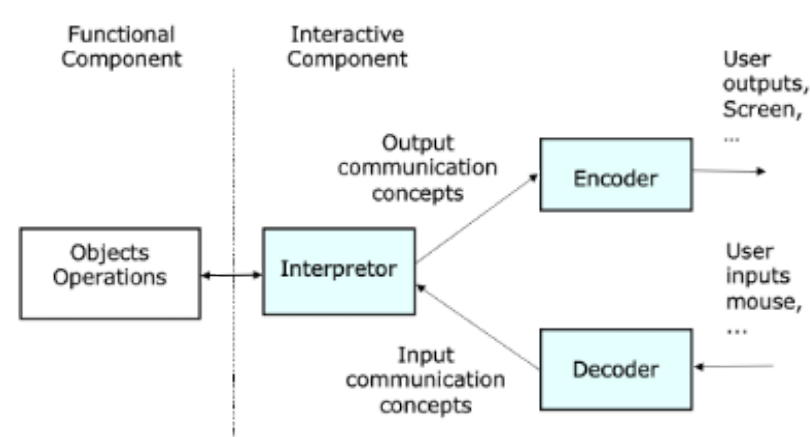
 [IMG_20210119_163221.jpg](#)

Question **4**

Correct

Mark 1.00 out of 1.00

The Output Communication concept could be:



Select one or more:

- ☒ a. Information to reset a value in Encoder ✓
- ☐ b. Information to add two numbers in Encoder
- ☐ c. Information to subtract two numbers
- ☒ d. Information to set up a value in Encoder ✓

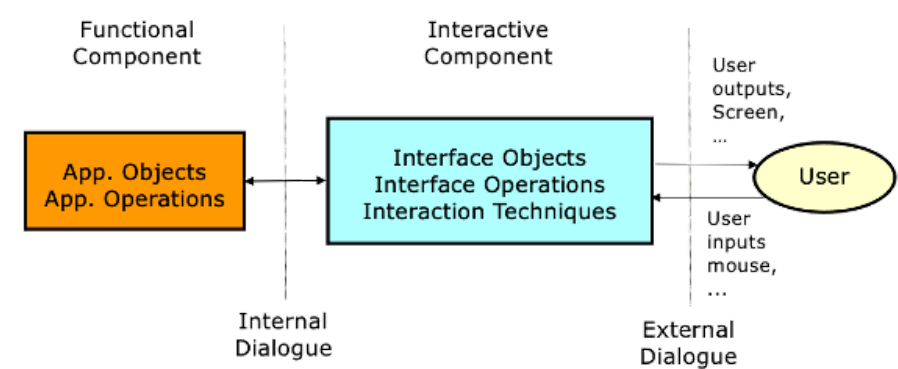
Your answer is correct.

Question **5**

Partially correct

Mark 0.50 out of 1.00

Dialogue Independence concept concerns with:



Select one or more:

- ☒ a. Separate the functional and interactive components ✓
- ☒ b. External dialogue ✗
- ☒ c. Internal dialogue ✓
- ☐ d. Interaction style

Your answer is partially correct.

You have selected too many options.

Question **6**

Incorrect

Mark 0.00 out of 1.00

The Formative Evaluation of the usability that check the correctness of the tasks on the low fidelity prototype is:

Select one:

- ☒ a. Heuristic evaluation ✗
- ☐ b. Review based evaluation
- ☐ c. Model based evaluation
- ☐ d. Cognitive walkthrough

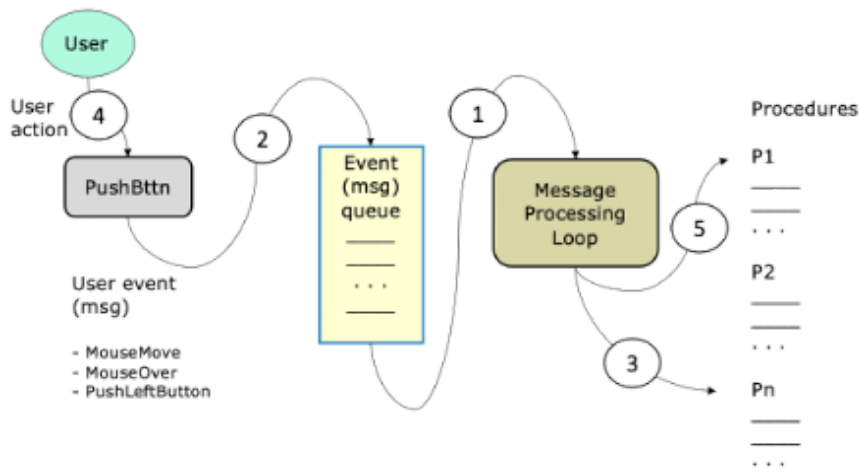
Your answer is incorrect.

Question **7**

Incorrect

Mark 0.00 out of 1.00

The event oriented flow control sequence throughout the following architecture is:



Select one:

- ☐ a. 4, 2, 1, 5, 3
- ☒ b. 4, 2, 5, 1, 3 ✖
- ☐ c. 1, 2, 3, 4, 5
- ☐ d. 3, 5, 1, 4, 2

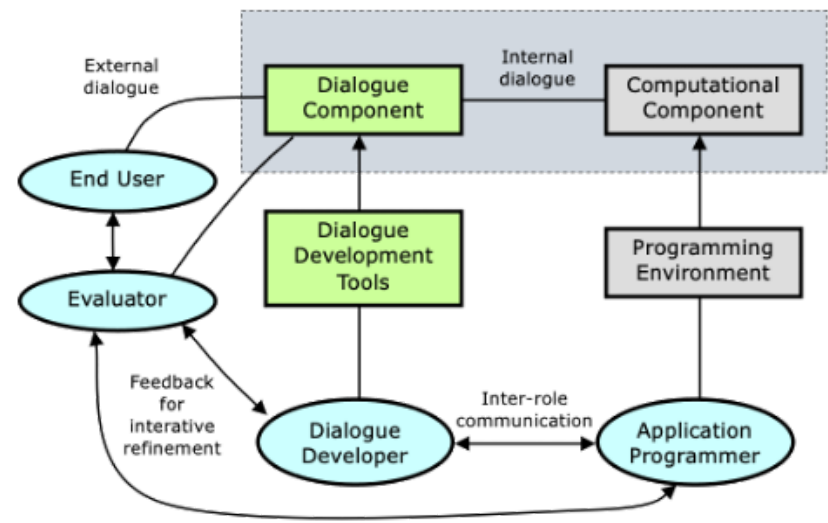
Your answer is incorrect.

Question **8**

Partially correct

Mark 0.50 out of 1.00

The following schema describes:



Select one or more:

- ☐ a. Dialogue Independence based software development methodology
- ☐ b. Evaluation methodology of the software development team members
- ☒ c. User Interface Management Systems methodology ✔
- ☐ d. Voice based internal dialogue within a software team

Your answer is partially correct.

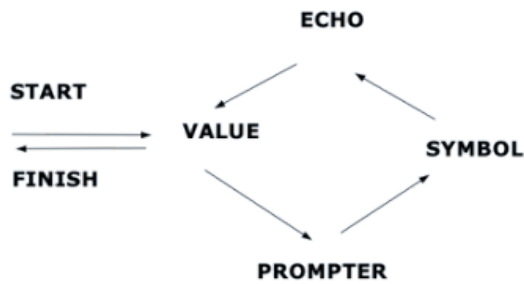
You have correctly selected 1.

Question **9**

Partially correct

Mark 0.50 out of 1.00

The following dialogue cell concerns with:



Select one or more:

- ☒ a. Nonlinguistic user interaction cycle ✓
- ☐ b. User interaction cycle associated to a push button
- ☐ c. Incremental prototyping of an interactive code sequence
- ☐ d. Incremental component development methodology

Your answer is partially correct.
You have correctly selected 1.

Question **10**

Correct

Mark 1.00 out of 1.00

Rapid prototyping concept could be classified as:

Select one or more:

- ☐ a. Programmable
- ☒ b. Revolutionary ✓
- ☒ c. Intermitent ✓
- ☐ d. Economical

Your answer is correct.

Question **11**

Complete

Marked out of 5.00

Exemplify and explain the Invocation Communication Concept as implicit and explicit specifications for commands of various syntactical forms.
Give examples of 3D Graphics editing.

Input communication concept contains:

- input gathering
- Invocation
- execution
- termination

Invocation communication concept:

- it used to invoke operations
- it uses states :active or inactive, execution or waiting and variable that helps control the states: active and execute

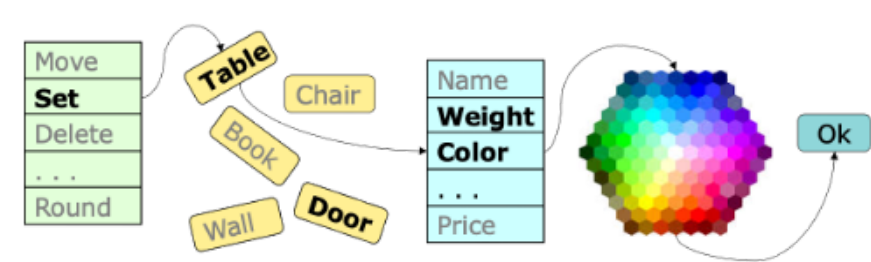
The invocation communication concepts can be interpret by some requests: execute, activate, preview.

Question **12**

Correct

Mark 1.00 out of 1.00

What is the knowledge encapsulated in GUI components in order to support the following possible user inputs?



Select one or more:

- ☒ a. Basic predicate ✓
- ☐ b. Interface object attributes
- ☐ c. User experience
- ☒ d. Input gathering predicate ✓

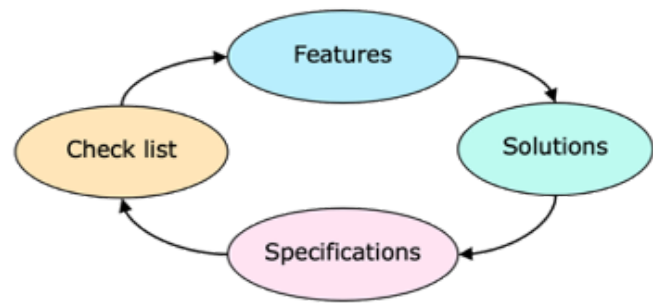
Your answer is correct.

Question **13**

Complete

Marked out of 5.00

Exemplify and explain the Usability Design Process presented in the figure for the Simplicity and Feedback usability principles, in the GUI of a 3D Graphical Editor. Detail each phase and highlight the main issues of such designing approach.



Usability measures how well a user in a specific context can use a design to achieve his/hers goal efficientlu. Designers measure a design's usability using a development process toensure maximum usability.

Check list: - we see all the features that can help us develop the application

Features:

- Simplicity principle :

This principle makes simple task simple to execute. It provides shortcuts for the user and also communicates clearly and simply.

- Feedback principle:

Feedback keeps the user inform about all the changes, states, actions and interpretations, errors and so ok. It uses a concise language for the user to understand clearly what is the state of the app. Contains all the information and data gathered in evaluations and the user feedback given during the development.

Solutions & Specification:

I will try to use the same components in the different scenarios in order to simplify the user experience.

An example is shown in the picture.

 [IMG 20210119 173113.jpg](#)

Question **14**

Correct

Mark 1.00 out of 1.00

Learnability attribute of usability may be improved by:

- Select one or more:
- ☒ a. Predictability ✓
 - ☒ b. Generalization ✓
 - ☐ c. Task completeness
 - ☐ d. Observe the internal state

Your answer is correct.

Question **15**

Correct

Mark 1.00 out of 1.00

What data gathering technique could be directed to a consensus view and could highlight areas of conflict?

- Select one:
- ☐ a. Individual interview
 - ☐ b. Naturalistic observation
 - ☒ c. Group interview ✓
 - ☐ d. Questionnaires

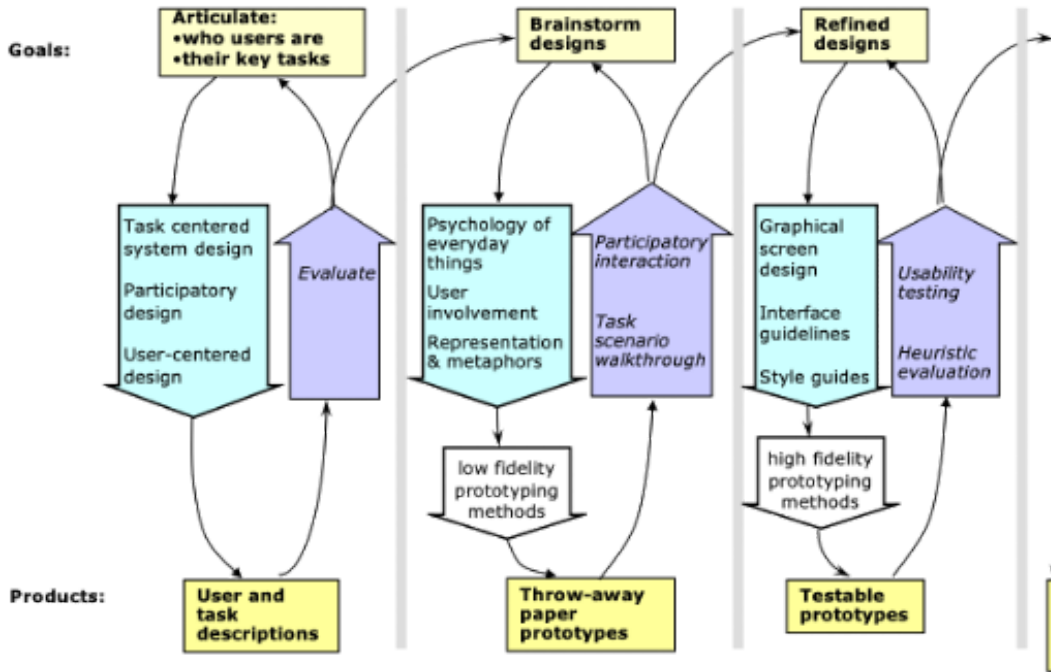
Your answer is correct.

Question **16**

Incorrect

Mark 0.00 out of 1.00

What are the two main concepts throughout the Calgary UID Methodology:



- Select one or more:
- ☒ a. Task formal description ✗
 - ☒ b. Prototyping ✓
 - ☐ c. Phase order flexibility
 - ☐ d. User and task description

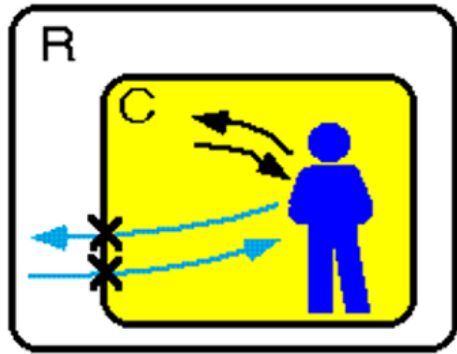
Your answer is incorrect.

Question **17**

Correct

Mark 1.00 out of 1.00

The following figure describe the concept of:



Select one:

- ☒ a. Virtual Reality User Interface (VRUI) ✓
- ☐ b. Graphical User Interface (GUI)
- ☐ c. Ubiquitous Computer User Interface (UCUI)
- ☐ d. Augmented Reality User Interface (ARUI)

Your answer is correct.

Question **18**

Correct

Mark 1.00 out of 1.00

In which phase of the WebQEM method is weighted the importance of elementary preferences?

Select one:

- ☐ a. Analysis and documentation of results, conclusions and recommendations
- ☐ b. Define and specify quality requirements
- ☒ c. Global evaluation ✓
- ☐ d. Elementary evaluation

Your answer is correct.

Question **19**

Correct

Mark 1.00 out of 1.00

The Output Communication model could be:

Select one or more:

- ☒ a. Continue refresh model ✓
- ☒ b. Message model ✓
- ☐ c. Graphical model
- ☐ d. Linguistic model

Your answer is correct.

Question **20**

Correct

Mark 1.00 out of 1.00

Fitts's Law based evaluation is performed to optimise the following feature of GUI:

Select one or more:

- ☐ a. The aspect of the interface objects
- ☒ b. User interaction time ✓
- ☐ c. The event oriented interaction cycle
- ☒ d. Movement of the cursor throughout the screen space ✓

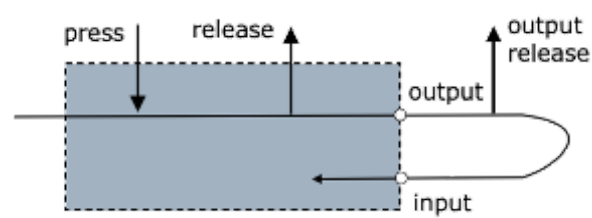
Your answer is correct.

Question **21**

Correct

Mark 1.00 out of 1.00

The following figure describes graphically the following gesture as a user input technique:



Select one:

- ☐ a. Press-timer gesture
- ☐ b. Drag gesture
- ☐ c. Range gesture
- ☒ d. Click gesture ✓

Your answer is correct.

Question **22**

Correct

Mark 1.00 out of 1.00

The following interaction technique is a gesture:

Select one or more:

- ☐ a. drag and drop
- ☐ b. combo box
- ☒ c. click ✓
- ☒ d. drag ✓

Your answer is correct.

Question **23**

Complete

Marked out of 5.00

Exemplify and explain the extension of some usability heuristics in order to discover and identify detailed usability problems. Exemplify the heuristic: Match Between System and Real World, for application GUI: controlling the movement of an avatar within a video game.

Heuristics are used to find usability problems. Heuristics make the evaluator exams the interface in order to make the design compliant and recognizable for the user.

These evaluation methods are now widely taught and practiced in the new media sector, where UIs are often designed in a short space of time on a budget that may restrict the amount of money available to provide for other types of interface testing.

Match between system and real world

The system should use a language simple for the user to undersand. The words, phrases and concepts should be familiar to the user and also the information should apper in a natural and logical order.

Before the design of the application is done, you should know the user because terms, concepts that are clear to may not be clear to the user too. To achieve the desired outcome, the designer has to follow the natural mapping. By doing this, the user will get familiar and will remember more easily the design of the application.

The exemplification is shown in the picture. The arrows in the picture represents the buttons.

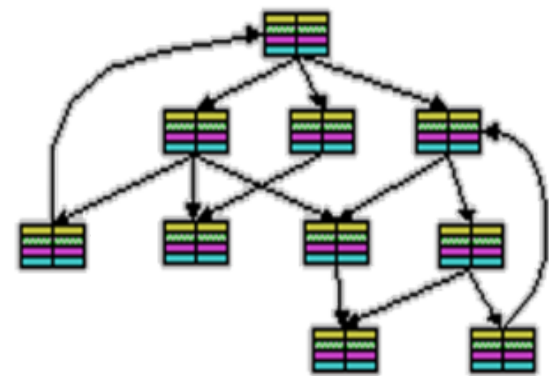


Question **24**

Correct

Mark 1.00 out of 1.00

The following figure describes graphically the following menu as a complex user input interaction technique:



- Select one:
- ☐ a. Linear sequence menu
 - ☐ b. Acyclic network menu
 - ☐ c. Tree structure menu
 - ☒ d. Cyclic network menu ✓

Your answer is correct.